University of Toronto Statistical Sciences Research Program 2026

Unofficial Project Report on Uncertainty Quantification for Predicting Stellar Properties from Gaia Spectra

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Contents

1 Introduction

The University of Toronto Statistical Sciences Research Program (UTSSRP) is a unique opportunity granted to 22 Canadian undergraduate students for the 2026 cohort, comprised by two intensive weeks of: advanced group research under a supervisor, graduate-level lectures and seminars on various statistical topics, and above all a valuable experience to learn and grow academically and professionally. Numerous esteemed faculty and graduate students from the University of Toronto (UofT) made time from their busy schedules to teach and support the undergraduates through our learning journey. A list of acknowledgements of the faculty, graduate students, and those personally involved in my overarching UTSSRP process (references, application, and support) is listed at the end of this paper.

The purpose of this paper is to produce a comprehensive summary of what content was present across the entirety of the UTSSRP. This summary includes: the lectures and seminars presented with my understanding and main takeaways from each, a recollection of events/phases of the research project I was fortunate to participate in, the findings and outcomes of the project, and finally an outline of the events the cohort had an opportunity to join in for and work-life balance.

The program was in its second iteration this year, running from Monday June 8th to Friday June 19th, on a 9:30AM-5:30PM weekday schedule officially; but as academics, we all very well know that work never stops at a certain time of day. The UTSSRP team was gracious enough to offer travel compensation for all students (including flights, VIA Rail, and GO train rides) along with shared residence units (private room within shared 4-bedroom unit) at Campus One, breakfast and lunch on weekdays, and certain organized events for work-life balance and networking opportunities. For the residence units, each unit housed students of the same gender and members part of up to two differing projects (thus 1:3 or 2:2 split amongst projects, or all 4 in the same project). Students were accepted and came from all over Canada, along with a few Canadian citizens/permanent residents from U.S.-based universities, with a good split between UofT, Ontario, and out-of-province undergraduates was observed. As likely inferred, the program was in-person on the University of Toronto Campus, on their St. George campus in the Department of Statistical Sciences office floor.

Note - Level of Reading

The paper will attempt to explain every topic in a way that is comprehendible by individuals with an exposure to undergraduate statistics, calculus, linear algebra, and machine learning terminologies—thus it is assumed that those reading have at least basic familiarity with such topics. Of course, not everyone reading this will have this knowledge given its breadth, or simply not having studied some of these topics. As such, if the reader ever would like any clarification, the author is available to contact via any means the reader already has capability of doing so, if any.

2 Program Overview

There were four main components to the program overall:

- Supervised research project in groups of 3-4 plus Ph.D student TA and Faculty Professor supervisor
- 2-hour introductory graduate-level lectures provided by Faculty members
- 1-hour introductory graduate-level seminars and panels by Faculty members
- After-hours events organized by the UTSSRP social committee

The first three components were all part of the day-to-day schedule, while events ran every few days after research concluded for the day and during the weekend to promote work-life balance. The first day had a brief campus orientation, while the last day replaced the research portion with lightning talk-style presentations of our research projects to Faculty and students. During lunches, Faculty members sat in and conversed with us in a group coffee chat style, and at times we had bonus lectures regarding professional development skills during the lunch hour. Breakfasts and lunches were catered and provided free of cost.

The full program schedule is shown in the three tables on the next few pages. The research time slot is the same for all groups, where we had our own rooms and desks for each group in the floor.

Week 1	Mon Jun 8	Tue Jun 9	Wed Jun 10	Thu Jun 11	Fri Jun 12
Lecture (9:30AM - 11:30AM)	Mark-Recapture Models by Prof. Gracia Dong	Robust Stat. Methods (Pt. 1) by Prof. Keith Knight	Pseudo/Quasi-Random Numbers by Prof. Gracia Dong	Robust Stat. Methods (Pt. 2) by Prof. Keith Knight	High-Dimensional Data by Prof. Elena Tuzhilina
Lunch (11:30AM - 1:30PM)	Bonus Lecture: Networking Development by Max Ma	—	—	—	—
Seminar (1:30PM - 2:30PM)	Orientation Campus Tour (no seminar)	Statistical Perspectives on the Replication Crisis by Prof. Jun Young Park	Beyond Classical Kernel Estimators by Prof. Thibault Randrianarisoa	Statistical Theory by Prof. Nancy Reid	Ph.D student presentations on various stat. topics
Research (2:30PM - 5:30PM)	Project outline and initial meeting; data definitions, EDA (Part 1)	Linear regression baseline; aleatoric/epistemic uncertainty (Part 1)	Model selection discussions; initial SBI model v1 (Part 2)	Hyperparameter grid search (80 runs); NSF vs. MAF exploration (Part 2)	Final architecture selection; model review with Prof. Speagle; UQ intro (Part 2)
Event (After 5:30PM)	Pub Night at Sin & Redemption	—	Ice Cream & Park Day at Trinity Bellwoods	—	—

Weekend	Sat Jun 13 – Sun Jun 14
Event	Toronto Islands Trip (Saturday, June 13). No lectures, seminars, or formal research slot this weekend.

Week 2	Mon Jun 15	Tue Jun 16	Wed Jun 17	Thu Jun 18	Fri Jun 19
Lecture (9:30AM - 11:30AM)	Mathematics of Generative Modelling (Pt. 1) by Prof. Ricardo Baptista	A Practical Introduction to Bayesian Inference by Prof. Joshua Speagle	Mathematics of Generative Modelling (Pt. 2) by Prof. Ricardo Baptista	Posterior Sampling and Posterior Predictive Checking by Prof. Joshua Speagle	Stochastic Differential Equations by Prof. Sebastian Jaimungal
Lunch (11:30AM - 1:30PM)	—	—	Bonus Lecture: Professional Development by Max Ma	—	—
Seminar (1:30PM - 2:30PM)	Career Panel by Mufan Li, Kevin Zhang, and Michael Moon	The Magic of Monte Carlo by Prof. Jeff Rosenthal	Spatiotemporal Statistical and ML Methods for Environmental Learning by Prof. Meredith Franklin	Risk Assessment Under Uncertainty and Ambiguity by Prof. Silvana Pesenti	Preparation time for poster session (no seminar)
Research (2:30PM - 5:30PM)	Uncertainty quantification: fudge factor methodology (Part 3)	Conditional coverage analysis; conformal prediction (Part 3)	Conformal prediction extensions; results consolidation (Part 3)	Poster creation (Part 4)	Lightning-talk poster session; closing ceremony
Event (After 5:30PM)	Board Game Night on-campus	—	—	FIFA Watch Party (Canada vs. Qatar)	Tilt Arcade Bar Celebratory Outing

The first week included an earlier start on Monday, June 8th at 8:30AM in order for a student networking session and program introduction presentation to be provided. All students arrived on campus on Sunday, June 7th in order to move into our CampusOne residence units.

Any research or learning done during the weekend was at the own behest and commitment by the students individually, as it typically follows in a regular university schedule while students would take courses.

3 Research Project: Uncertainty Quantification for Predicting Stellar Properties from Gaia Spectra

Under the applied project group, supervised by Prof. Joshua S. Speagle and TA Ian Zhang, over the course of 12 days me and my teammates (Umang Mehta, Johan Phan) were tasked with performing research to build machine-learning models that are able to utilize a specialized input called Gaia spectra to predict stellar properties. Gaia spectra are observations made by the Gaia satellite of the European Space Agency (ESA), which are a condensed form of APOGEE spectra, where the latter are high-resolution wavelength observations of many stars. Thus, Gaia spectra are a lower-resolution, but much more computationally cheaper form of wavelengths in BP and RP (blue and red) spectra, with a final dimension of 110 coefficients. As for the aforementioned stellar properties, these are:

- $T_e$ff: Effective temperature, or surface temperature of the star
- $log(g)$: Surface gravity of a star on a log-scale
- $[Fe/H]$: Metallicity, or the ratio between metals and non-metals in a star measured in units of dex, which can help deduce a record of a star's origin.

Definition - Metals in Astrophysics

Metals in the field of astrophysics are conservatively considered as any element that is heavier than helium, due to the abundance of hydrogen and helium throughout the universe relative to all other elements.

The property of highest interest to our group was that of **metallicity**, where through **Part 1** we will outline how this was the toughest stellar property to predict consistently.

To perform these predictions, the group first had a linear regression model built as a baseline, to better hypothesize of challenges/uncertainty that might be present within the data (**aleatoric uncertainty**) and to see what issues could arise from the simple model's performance (model performance having **epistemic uncertainty**). A more descriptive definition of both types of uncertainties will be provided later in the paper, along with a further explanation of results in **Part 1** (Section 3.1). Given the baseline, our group was also interested in creating more advanced models, in which we all were most interested in building with neural networks. Hence, each of us undergraduate students created three different neural network-based models that all had the same end goal: predicting metallicity values—however all with fundamentally different methodologies and architectures, of which they will be further described in **Part 2** (Section 3.2).

The original idea of the project was to utilize simpler methods, like KNN or Random Forests to compare with the baseline. However, through discussions amongst ourselves and our supervisors, we ultimately were most interested in building neural network models for learning experience, and given the natural extensibility of neural networks to a plethora of methodologies which include statistics-based expansions.

Finally, after all of this model-building and eventually tuning, came the most novel portion of our project. Uncertainty quantification fell into two parts: a manual approach using a **fudge factor** c which was chosen to tune our estimation interval widths to meet certain coverage levels, and an automatic approach using **conformal prediction** that uses a user-set scoring function to automatically adjust interval widths to meet coverage levels. Both of these approaches are further explored in **Part 3** (Section 3.3).

To conglomerate all of our findings and be able to present them in the poster session on the final day of the UTSSRP, we made an academic poster to showcase our project. This will be further spoken about in **Part 4** (Section 3.4).

3.1 Part 1: Exploratory Data Analysis, Baseline Formulation

The first two days of the research project were centred around introducing ourselves amongst TA and Professor, getting a sense of what our project will be all about, setting up our coding environments, and performing exploratory data analysis (EDA) alongside creating the baseline linear regression model. This entire process was not limited to only figuring out how to wrangle the given data, figuring out what standardization needed to be done, and plotting to find correlations amongst variables—but it was also used for understanding exactly *what* the data is since the physics underlying the Gaia spectra and how it's correlated with our target values, not to how mention the particular physics-based data standardization technique worked, was of great interest for each of us. Without having a full understanding of what it is our data does and how it works, the project would serve very little meaning and explainability of our findings would be inconsequential. Besides, we're working with stars, who wouldn't be at least a little curious as to how this all meshes together?

3.1.1 Day 1

Data Definitions and Standardization Methods

Prof. Speagle presented a slide deck to the group, outlining key details of the UTSSRP in general and then further introducing us to the overall gist of the project. As mentioned before, we are using Gaia spectra which are condensed, lower-resolution data procured from the APOGEE dataset, a high-resolution dataset. [1] The entire Gaia dataset holds ∼263,000 rows (stars) with 110 columns (spectral coefficients), thus we were working with a somewhat high-dimensional problem. The entire point of Day 1 was to perform simple EDA and create numerous plots which helped us visualize what is going on with the data. One important distinction is that for the remainder of the project, we only used a random sample of 30,000 stars in order to have more efficient computations throughout the notebooks.

From the Gaia spectra, of course we have the target values mentioned earlier: effective temperature, surface gravity, and metallicity. These are possible predictors from the spectra, which astronomers are most readily able to predict given stellar spectra.

To further define Gaia spectra, recall the aforementioned BP and RP terms. These are short-form for **Blue Photometer** (BP) and **Red Photometer** (RP), named for the corresponding prisms on the Gaia satellite which intake wavelengths from stellar objects and measure on the blue or red wavelengths respectively. Together, the two are called "XP". As mentioned, Gaia spectra are *compressed* into a form of special basis Hermite polynomial function, which can reproduce the full wavelengths if desired. This Hermite polynomial's specific definition was not relevant to our project discussion and findings. As mentioned, these spectra are comprised of 110 coefficients (XP), of which the first 55 are BP and latter 55 are RP coefficients. Later we will explore methods used to reduce these dimensions as desired.

Definition - Hermite Polynomials

Hermite polynomials are a special form of polynomial that satisfy certain properties, such as the differential equation $y'' - 2xy' + 2ny = 0$ and the recurrence relation $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$, and, most importantly, orthogonality on the real line. They are defined by a weight function e^{-x^2} and are used primarily in statistics with association to a Gaussian distribution, quantum mechanics, and numerical analysis.

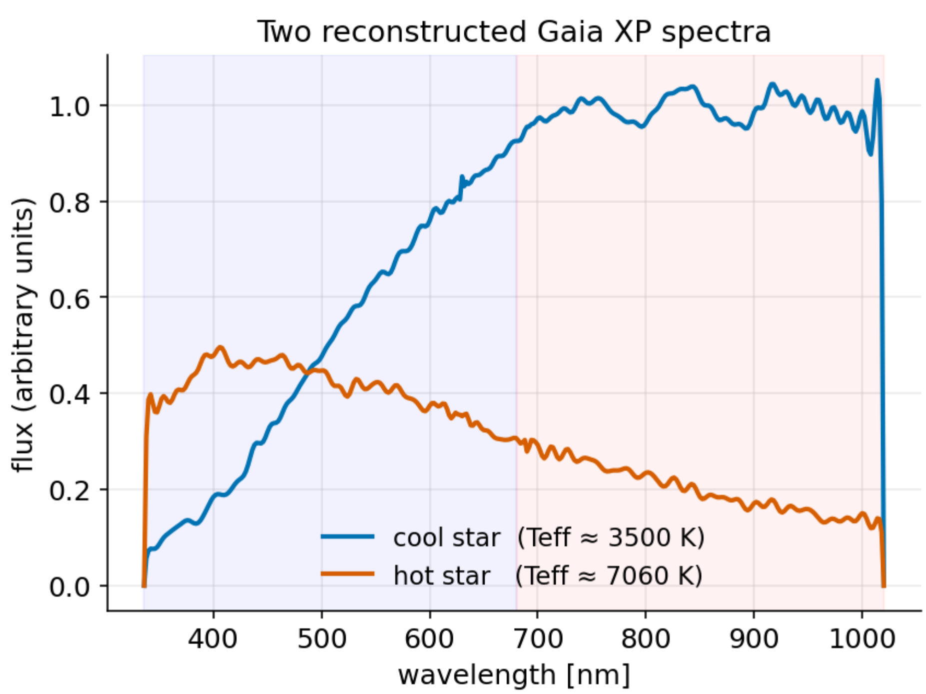

Figure 1: Plot of how two reconstructed wavelength spectra look when the Hermite basis polynomial is applied on the Gaia spectra for an arbitrary cool star and arbitrary hot star in the dataset. Each point on the lines represents one coefficient, re-scaled to the

corresponding wavelength.

As the reader might attentively recognize in the above figure, the y-axis is a measure of **flux** in arbitrary units. This comes from the value of what's known as **gflux** from the dataset. Gflux is a measure of Gaia broad optical brightness; in essence, the intensity of the brightness relative to our observation point, the Gaia satellite. The un-normalized spectra are subject to numerous factors that lead to highly differing relative brightnesses for different stars: distance, stellar dust, gravitational fields, and others affect how lightwaves are viewed. Thus, by normalizing by dividing out gflux, we set every star's relative spectra shape to be on an equivalent scale making predictions much more feasible. We keep the spectral shape because that is what drives the target values. Normalizing in this fashion is known to us as a **physics-based normalization**, by cancelling out any relative intensity/brightness differences present due to the aforementioned factors.

We also had the choice of performing a **generic standardization**, wherein each Gaia spectra is transformed to have zero mean, unit variance. This could be performed since some of the 110 Gaia coefficients are on a scale 4 orders of magnitude larger than others, thereby overpowering the overall Gaia spectra data. This would break any methods with distance-based measurements, like KNN.

Later, the paper will examine what choices led to us using either of the normalization/standardization methods.

EDA Plots and Dimensionality Mapping Techniques

In this segment, we will explore a few plots the group made to further explore the data, along with a short analysis on how two dimensionality reduction techniques (PCA for linear reduction, and UMAP for non-linear reduction). The tasks were split into different parts with different students leading the sections, along with the dimensionality reduction techniques being done by all the students collaboratively.

Umang was responsible for creating plots showcasing how temperature and metallicity of stars varies given spectra, along with signal-to-noise ratio and error structure plots. Johan was responsible for identifying which coefficients were the most explainable for each target value (T_eff, $log(g)$, $[Fe/H]$). I was responsible for finding label distributions and pair-wise relationships amongst the target variables and the input data, along with producing a Kiel diagram to show what differences in star classes exist within the data.

Definition - Kiel Diagram

A Kiel diagram is a specialized version of the Hertzsprung-Russell (H-R) diagram, both of which are plots used in astronomy. The Kiel diagram plots $\log(g)$ against T_eff to show the difference amongst star classes (dwarfs, giants) present in a dataset.

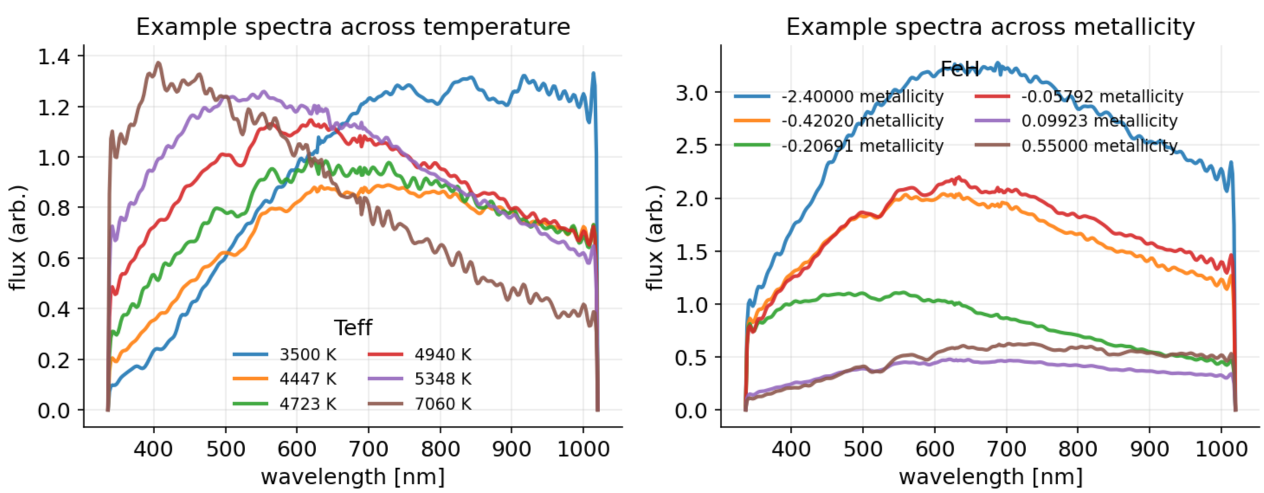

Figure 2: For spectra across temperature, there is a fairly clear distinction between where the intensity of a certain wavelength is strongest for a very warm star versus a very cool star. The left plot shows that the warmer stars have highest average spectra intensity in the lower wavelengths, while cooler stars have highest average spectra intensity in the higher wavelengths. Meanwhile, across metallicity, there is no clear trend between spectra intensities and metallicities.

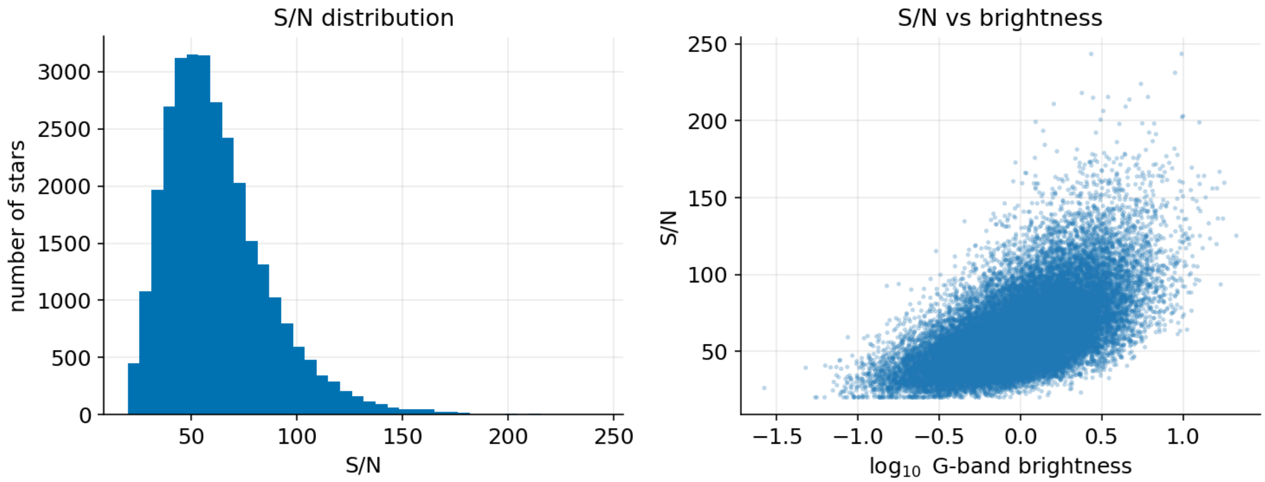

Figure 3: While signal-to-noise ratio is a relative measure, it does show that the majority of stars fall under a relatively explainable portion, with a few outliers in favour of being *more* explainable with a higher signal, shown in the left plot. The plot on the right shows how explainability varies across brightnesses, with brighter stars having a stronger signal. This correlates with particular coefficients carrying the brightness and intensity features, which are further deduced below.

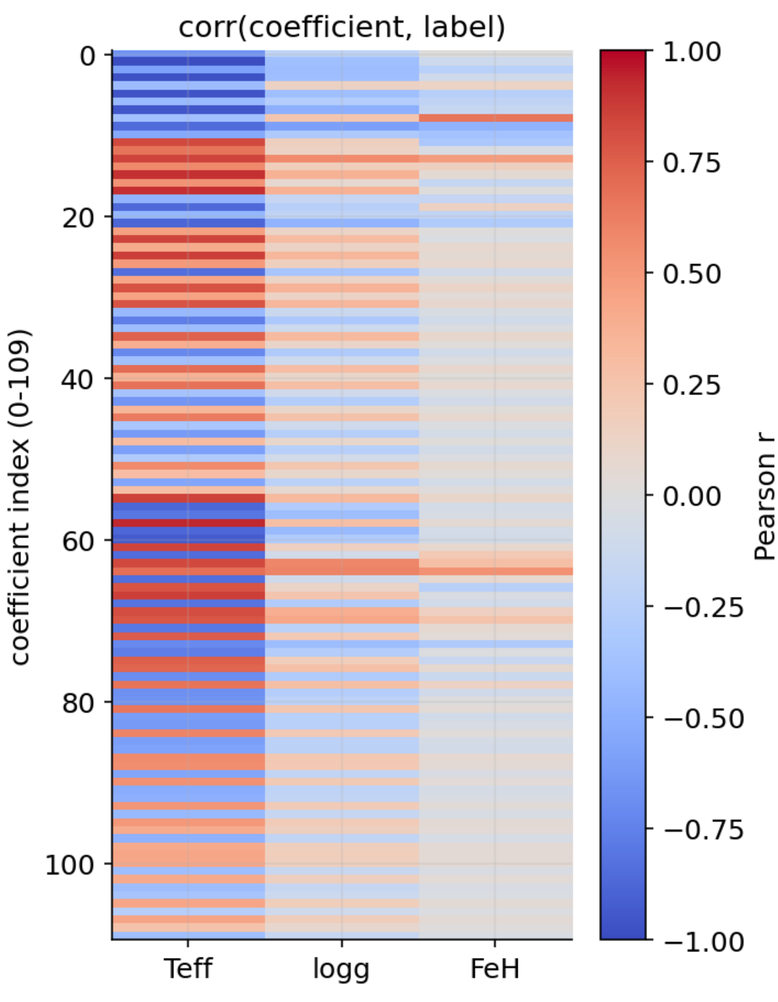

Figure 4: For effective temperature, there is a very wide variety of spectra coefficients that are indicative of a star's temperature, making it easy to predict. Slightly less-so with surface gravity, but still relatively explainable. Whereas with metallicity, very few coefficients hold statistically relevant explainability for predicting it. The top 5 coefficients for explaining metallicity, were 8 (BP), 64 (RP), 13 (BP), 9 (BP), and 10 (BP); each with Pearson r values of 0.66, 0.54, 0.48, -0.47, and -0.36 respectively; four of the five coefficients are BP coefficients.

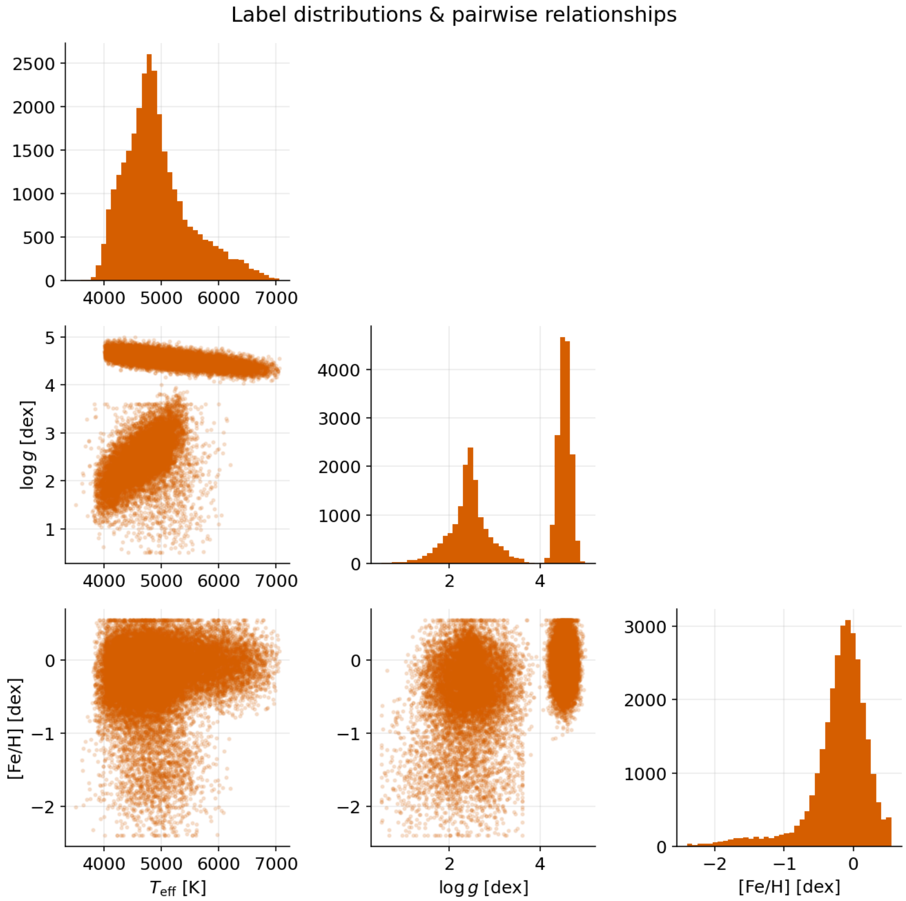

Figure 5: The most intuitive finding from this plot is the bimodal distribution on $log(g)$, clearly showing that there are two subgroups of stars within the dataset: the lighter surface gravity giants, and the heavier surface gravity dwarves. Meanwhile metallicity has a large amount of stars around a similar level to that of our Sun, represented by 0.0dex, and a tail of stars with lower metallicities. Effective temperature is set around 5000 K for the majority of stars. The pairwise relationships show minimal correlation between temperature and metallicity, and between surface gravity and metallicity. See the next figure on the analysis of the pairwise relationship between effective temperature and surface gravity.

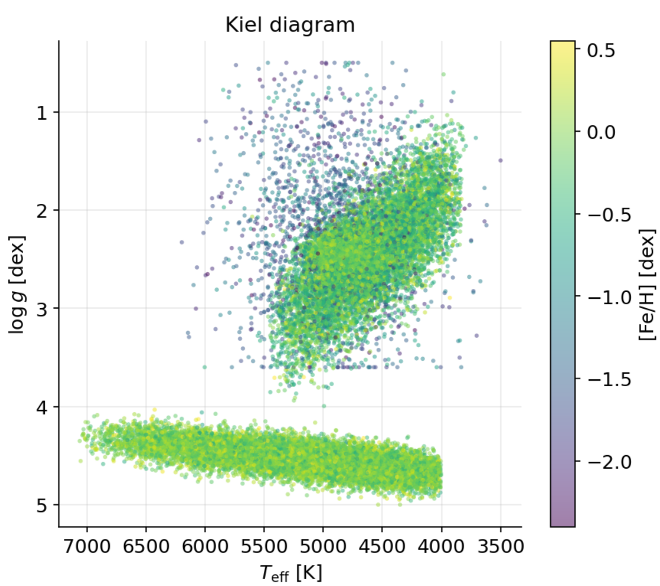

Figure 6: This Kiel diagram plots the pairwise relationship between effective temperature and surface gravity. The higher the surface gravity, the broader in range the surface temperature can be, however the majority of these dwarves have a Sun-like metallicity. Meanwhile the lower surface gravity stars are quite set on a range of higher effective temperatures, with metallicity varying, but typically being lower than that of the Sun.

After clarifying numerous questions with Prof. Josh and TA Ian regarding the data, we then moved on to analyzing dimensionality reduction with linear (PCA) and non-linear (UMAP) methods.

For PCA, we identified that reducing components down from 110 full spectra to 4 principal components via visual analysis of a scree plot could potentially benefit our models with simpler data. Later, we will note what models were created and whether PCA reduction helped or not. An interesting find was that to reach 90% of cumulative explained variance, we would need to utilize 54 principal components, which is already a great reduction as well. This is due to an observation that the majority of neighbouring coefficients are near-duplicates, and thus PCA is able to remove about half of these components in order to preserve around 90% of the variance. The 4 components proposed by the scree plot elbow explained $\sim 60\%$ of the cumulative explained variance.

As for UMAP, which is a slower reduction technique compared to PCA, it attempts to preserve which stars are close to each other distance-wise relative to the 110 Gaia coefficients, but it also maps the 110 dimensions down to 2. This is mostly done for visualization in our work and we did not end up exploring it further given that within the below figure, there is no clear correlation between the two UMAP axes and metallicity.

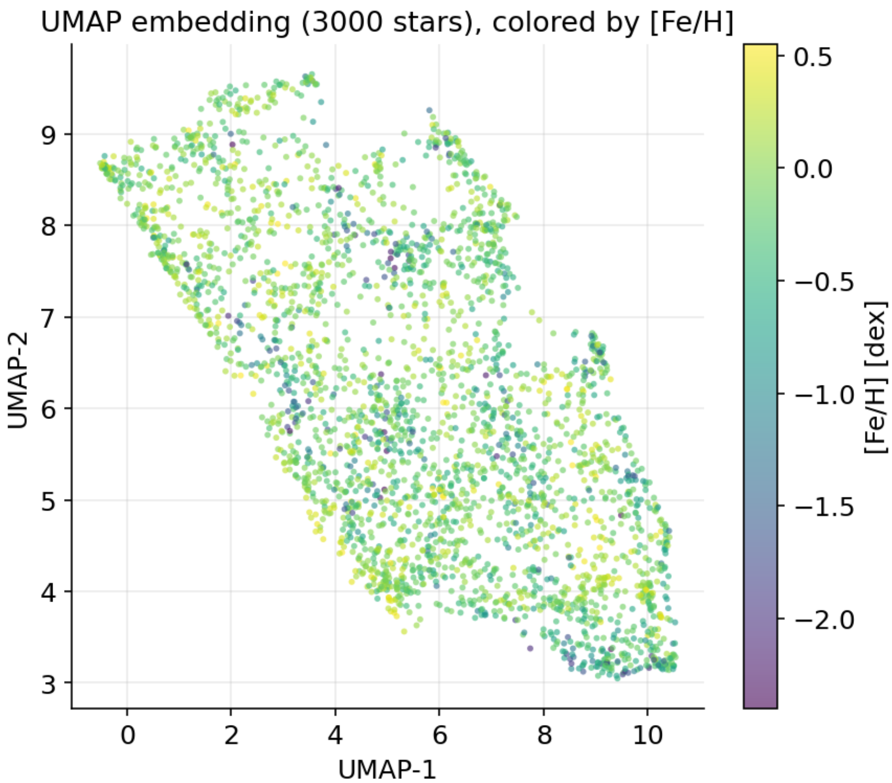

Figure 7: The found UMAP embedding, which doesn't show any clear correlation between its axes and metallicities.

We absolutely may have explored this technique further, but with time constraints and interest in different, more pertinent work for our project (model building, uncertainty quantification), we called it a day and prepared ourselves for further analysis and creating the linear regression baseline in Day 2.

3.1.2 Day 2

This day was focused on producing the linear regression baseline, and interpreting the results in order to gain a further intuition for the underlying data, and beginning to comprehend what **aleatoric** and **epistemic** uncertainty are.

Definition - Aleatoric and Epistemic Uncertainties

Aleatoric uncertainty is uncertainty present within the data itself, such as noise that cannot be removed. This type of uncertainty also cannot be removed from a statistical model by adding more of the data to the model. In our Gaia spectra, this kind of uncertainty can surface from photon noise, and uninformative metal lines in stars.

Epistemic uncertainty is uncertainty present within the model rather than in the data. This type of uncertainty is able to be mitigated through improving the hyperparameters of the model, or by adding more data to the dataset. It usually surfaces from uneven training datasets, and is largest when training samples are sparse.

In Day 1, we recognized, especially from the coefficient correlation plots made by Johan, that effective temperature and surface gravity are very well described by Gaia spectra. However, metallicity remains a difficult property to deduce. A first attempt by linear regression shows that despite the difficulty, a large amount of the data is able to get surprisingly strong metrics. Trained against physics-normalized Gaia spectra, the linear baseline found an RMSE of 0.198, MAE of 0.126, and most interestingly, an R^2 of 0.788. Since we did not know if our RMSE or MAE scores are relatively good, given they are arbitrary metrics that depend on running models numerous times, the explained variance of $\sim 79\%$ stood out to us. When a simple linear regression model is able to take a somewhat high-dimensional input and explain this much variance for a seemingly difficult-to-predict (low correlation amongst coefficients) target, then something could be present in the data that is able to explain the target to a strong degree even if initial insights showed otherwise.

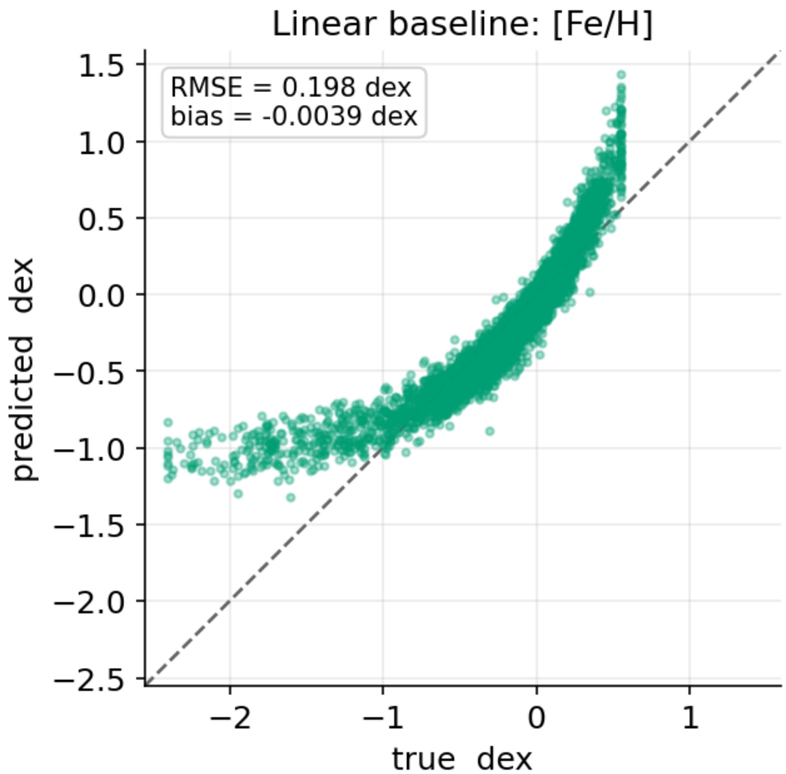

Figure 8: Our simple linear estimator is underestimating on average, mostly seen at the extremes of the metallicity range. Interestingly, the residuals seem to follow a normal-like distribution pattern, with the majority of the residuals lying visually close to the estimator, however the further towards the "tails" the data lies, the greater the error. In Section 3.2, the reader should pay attention to how Johan took advantage of this fact in order to create the Gaussian Predictor model and see the possible link between these residuals and the Gaussian assumption.

We then attempted to explore this by first plotting what coefficients drive feature importance and found that the first few coefficients in both BP and RP hold the most information related to $[Fe/H]$dex predictions.

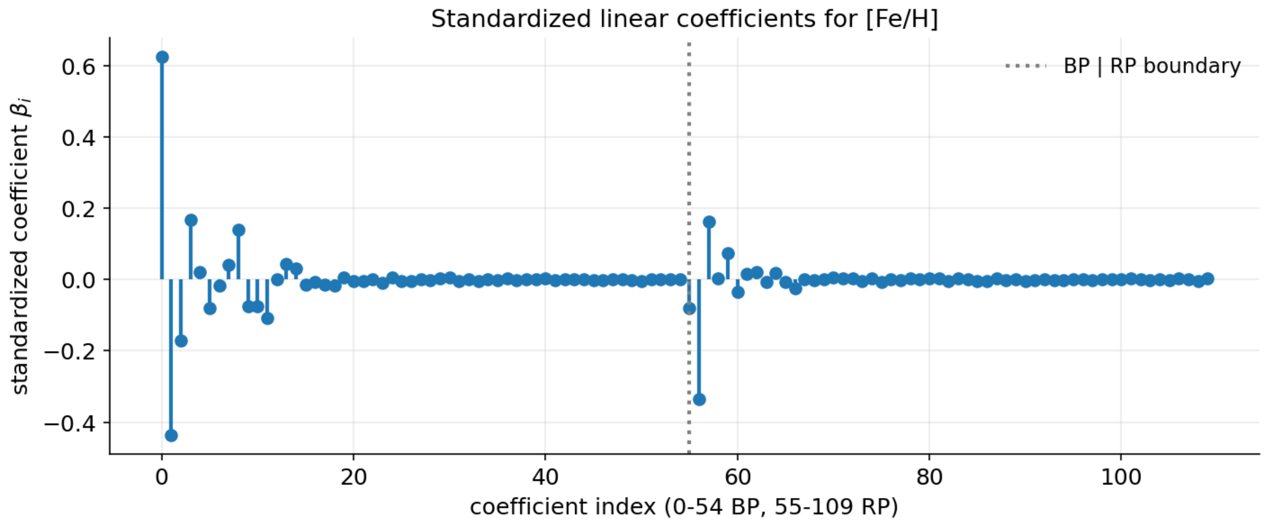

Figure 9: The first few coefficients in BP had the highest overall feature importance, and just a few in RP had some importance which was also in milder scale compared to BP. More on the numerical analysis below.

Then, we found what fractional contribution each coefficient outlines. The calculation for fractional contribution utilizes effect, which is defined as:

$$\text{effect}_i = \beta_i x_i$$

for a weight β_i applied to each observation (spectra) x_i during model training. Recall that model training for a linear regression model fits:

$$\begin{aligned} \hat{y} &= \beta_0 + \sum_i \beta_i x_i \\ &= \beta_0 + \sum_i \text{effect}_i \end{aligned}$$

for a bias/intercept term β_0. Given this effect, the formula for fractional contribution is:

$$\text{contribution}_i = \frac{|\text{effect}_i|}{\sum_j |\text{effect}_j|}$$

From this analysis, the following 5 coefficients were identified as having the largest fractional contribution for an arbitrary star:

Coefficient Index	Fractional Contribution	Percentage
0 (BP)	0.378	37.8%
56 (RP)	0.139	13.9%
8 (BP)	0.105	10.5%
1 (BP)	0.074	7.4%
2 (BP)	0.068	6.8%

Table 1: Top-5 XP spectral coefficients by fractional contribution to the model.

Recall that RP spectra begin at the 55th index (utilizing a 0 index in the above table, meaning the 55th index is the 56th coefficient). As such, the only Gaia coefficient in RP that is in the top 5 is coefficient 57, or the 2nd RP coefficient. The remainder are all early coefficients in the BP spectrum. Thus, the highest amount of explainability overall lies in BP (for an arbitrary star picked from the dataset). This method is simple, and was left as an exercise to us to create to gain an intuition for a method that *can* be used to find high-correlation features.

Another, more popular, method used was SHapley Additive exPlanations (SHAP), which is model-agnostic, defined by the following formula:

$$\phi_i = \beta_i(x_i - \bar{x}_i)$$

for training mean $\bar{x}_i$. Given that we standardized on our training dataset, $\bar{x}_i \approx 0$ meaning that $\phi_i \approx \beta_i x_i$, which is approximately effect$_i$. Thus, we effectively ran SHAP-for-linear in the earlier derivation of fractional contribution. In general, SHAP is powerful in non-linear cases, which is much more relevant in our later work where we will discuss non-linear methods of performing these predictions. We also performed an OLS summary given the calculated SHAP values, but did not explore this avenue greatly as we were beginning to formulate ideas for our Day 3 work, and figuring out how our error residuals look like.

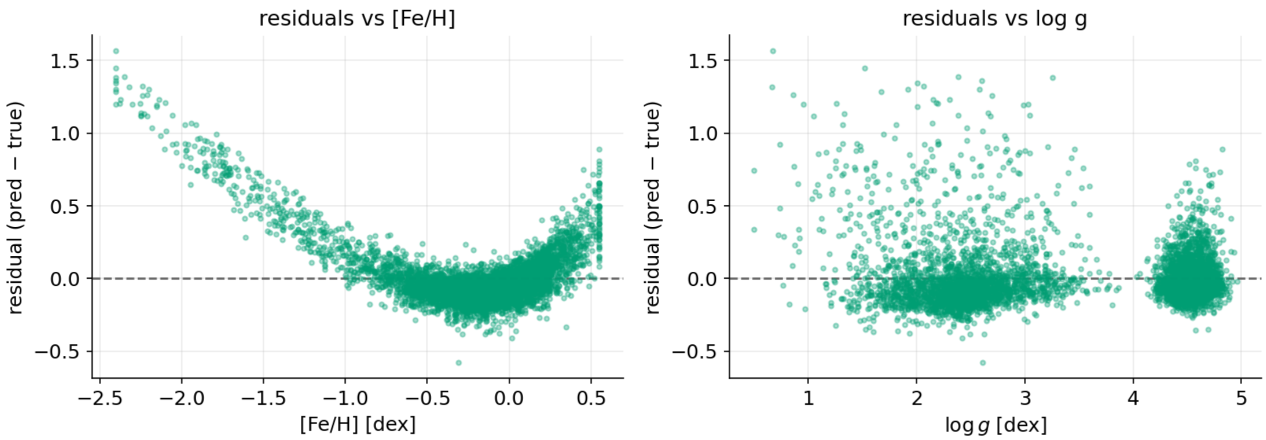

Figure 10: Note the lack of residuals showing low error between the [-0.5, 0.0]dex range in the left plot, and the bimodal scatter of $log(g)$ residuals in the right plot. More analysis below.

As the reader may see, there is a strong region where errors are quite low, around the [-0.5, 0.0]dex range. As the metallicity becomes lower and higher (metal-poor, to metal-rich), the residuals spike in a consistent manner. We also plotted the residuals for $log(g)$, and saw a bimodal scatter of the residuals. We deduced that these regions represent dwarfs (higher $log(g)$) and giant stars (lower). The higher the residuals, the more uncertain our model is (epistemic uncertainty), because the data is less informative (aleatoric uncertainty). Through the graphs, we may also note that it is particularly giant stars and metal-poor stars that will be the most difficult to predict upon accurately given their inherent aleatoric uncertainty.

Now, with all of this EDA in mind, we began exploring how to create our own models—an endeavour that took course over the next few days.

3.2 Part 2: Neural Network Model Tuning

As mentioned in the introductory segment of Section 3, the original idea of the project was to compare the linear regression baseline to simpler (relative to our final solution) non-linear algorithms like KNN clustering, Random Forests, and basic neural networks. Given all of our inclination towards building neural network models, through some discussions with Prof. Speagle and TA Zhang we collectively decided to instead pursue three different types of neural network solutions.

Team Member	**Model Type**	**Outputs**
Umang	Ensemble of Neural Networks	Predicted value of $[Fe/H]$dex
Johan	Gaussian Predictor	Mean μ and variance σ of approximated Gaussian
Edward	Simulation-Based Inference	Approx. posterior $p(\theta_i\|x_i)$ for input x_i, parameter θ_i

Table 2: Summary of each model that each team member built

The Ensemble model was comprised of numerous PyTorch neural networks that each were of varying depths and widths, but used the same inputs, activation, optimizer, and loss. After each network outputs its predicted value, the average of each prediction is taken to produce the final predicted value of $[Fe/H]$dex.

The Gaussian Predictor model was comprised of a singular neural network, which had two output nodes trained to provide the estimated mean and variance on a Gaussian distribution of the respective input data. The mean μ_i is the expected value of $[Fe/H]$dex for a particular input of Gaia coefficients x_i, and the variance σ_i is a simple measure of how uncertain the model is in finding this μ_i, as a higher variance means a higher standard deviation, and the standard deviation dictates how spread-out the data is across the Gaussian. The assumption of the output following a Gaussian distribution was made through some data exploration and intellectual judgment made on Johan's end, which will later be shown to be a good assumption.

The Simulation-Based Inference (SBI) model was substantially more involved in terms of architecture, as it attempted to combine Bayesian inference via **normalizing flows** with an **embedded neural network**. The particular SBI network we used was a **Neural Posterior Estimator (NPE)**, where other types of SBI networks are neural likelihood estimators (NLEs) and neural ratio estimators. For our case with NPEs, Bayesian inference assumes that there will be a prior distribution on the beliefs present in the data, and a posterior distribution that is desired to be approximated (only approximated due to complexity of directly finding and sampling from the theoretical distribution). **Normalizing flows** are a method used to transform a distribution (in this case, the base approximated posterior) by shifting, skewing, and scaling it given a set of parameters Θ that are learned from an **embedded neural network**. [2]

This neural network *typically* takes in simulated data from the prior distribution, where the prior's assumption/belief of how the target value ($[Fe/H]$dex) is distributed is sampled from, and matched with simulated Gaia coefficients created via mathematical models the model-maker creates, producing a (θ, x) pairing. Since we already had this pairing by having labelled data $\theta_i = [Fe/H]\text{dex}_i$ and x_i as the associated Gaia spectra—with both values corresponding to the i'th star in the dataset—the need for creating simulated data was by-passed. Instead, the prior was solely used to provide a possible range of values that $\theta = [Fe/H]$dex can take on, which is also the limit for any predictions that are done during training and beyond. In our case, we decided to keep the bounds as $[min([Fe/H]\text{dex}), max([Fe/H]\text{dex})]$ since we planned to use a holdout validation dataset rather than creating a testable model on entirely new data (out of scope for our project).

Note - SBI General Workflows with a Simulation Prior

In general, SBI generally has a simulator that a prior distribution is placed upon, based on a belief the researcher/user assumes about the input data they want to simulate. This particular distribution can be as simple as a uniform distribution, or as complex as needed; as long as it can be sampled from, that is the only important factor. For our use case, we also needed to ensure that data was **exchangeable** due to the assumption that **conformal prediction**, later defined, has in place to perform uncertainty quantification.

That aside, SBI then samples from this prior simulator in order to produce the (θ, x) pairing necessary to train the embedded network. The simulator takes in these parameters θ, or more commonly representing existing knowledge or constraints on your target posterior distribution, in order to produce the simulated observations x to pair with, which the embedded network then uses to train itself in a forward-facing fashion. Thus, the prior distribution is an independent, black-box component of an SBI model, which allows for ease of complete detachment and replacement if appropriately used, or more commonly to run numerous simulators in parallel to feed into an embedding network. The remaining steps are identical to what was outlined for our research use-case.

3.2.1 Day 3

With the context in mind, the next few days will have some rather short explanations as creating our networks entailed many unrecorded discussions and questions amongst the group. Day 3 entailed our second meeting with Prof. Speagle to finalize the choices of models we'd like to build. Ideally, we had planned to wrap up these neural network models and then also have some analysis on the simpler models originally planned to be worked on. However, as the days went by, we became much more entranced by our current solutions, and each of us having to learn new ways of leveraging neural networks (and for Umang, learning how they work entirely from scratch as he had not yet worked with them before).

Each of us had set out to find papers or other resources to educate ourselves on each topic, figuring out how to conjure a working model. In Johan's case, he had instead been exploring the data much further this day and intuitively producing the assumptions outlined in Section 3.2 for his end Gaussian Predictor model.

Near the end of the day, the SBI model had been created with a, to be completely candid, nearly-too-basic understanding of how the entire framework actually functions. A few assumptions were made by my part that seemed intuitive given the research done and the questions asked, and they were correct to an extent. Nonetheless, later in Day 5, a lesson that was learned from this is outlined.

Immediate noticeable improvement was shown with the initial SBI model, which had a very simple 1 hidden layer, 16 neuron architecture for the embedding network, using an NSF normalizing flow algorithm and 4 principal components as inputs x. Normalizing flow algorithms will be better defined in Day 4.

Model	**Test RMSE**	**Test R^2**
Linear Baseline	0.198	0.788
Initial SBI Model	0.158	0.864

Table 3: Comparison of Linear Baseline vs. basic SBI Model key metrics

3.2.2 Day 4

This day was filled with further exploration on each of our ends, learning further through some more resources, and shaping our solutions a little bit further slowly but surely. While Umang and Johan took an approach to simply learn about their models in statistical manners and not delve into the many specifics of hyper-parameter tuning their networks and data inputs, my approach was to test out many different configurations and find the best one that my model works with and understand why it would work given such parameters compared to the other ones. More brute-force, but such is the data scientists' life at times.

The following are the parameters and tested values upon which a grid search was performed for a total of 80 hyper-parameter tuning jobs, which ranged from running for 12 to 578 epochs:

- Hidden Layer Architectures: (4, 1), (32, 1), (64, 1), (110, 1), (4, 4, 1), (32, 32, 1), (64, 32, 1), (64, 64, 1), (110, 64, 1), (110, 110, 1)
- Input Dimensions: 4 (physics-based normalization + PCA), 110 (physics-based norm + PCA)
- Activations: ReLU, LeakyReLU(negative_slope=0.01)
- Normalizing Flow Algorithms: NSF, MAF
- Learning Rate: 5×10^{-4}
- Batch Size: 64
- Optimizer: Adam

A keen reader will question why PCA was done upon the input dimension of 110 inputs to produce a full 110 principal components. This choice was done purely out of interest to see what happens when the dimension isn't reduced, but the correlation is completely removed between each of the components through diagonalizing the covariance matrix that the data space lives in, and by subtracting the mean of the spectra from each coefficient (both key effects of PCA, excluding the dimensionality reduction in this case). A few model runs prior to hyper-parameter tuning curiously showed that through (accidental) running of PCA to produce 110 principal components produced better test RMSE and R^2 values on the same architectures where the pure 100 coefficients are placed through. Unfortunately, there was little time for discussion on this topic and was merely a curious by-product of something that was recognized during the model-tuning phase, and was not mathematically analyzed thoroughly as to why this had happened. However, perhaps this is an exercise for myself and the reader to think about why this might occur.

Through those 80 grid-search runs, the initial standout result was the following, before reviewing and evolving going into Day 5.

Input	**Depth**	**Width**	**Act.**	**Flow**	**LR**	**Opt.**	**Batch**	**Epochs**	**Test RMSE**
PCA 110	3	4, 4, 1	ReLU	MAF	5×10^{-4}	Adam	64	51	0.1027

Table 4: Best grid-search architecture from Day 4, prior to further refinement in Day 5.

One key thing to note is how the different normalizing flow algorithms work, as they aim to transform the parameters of the output distribution in different methods. The following block defines their methods conceptually.

Definition - NSF and MAF

Neural Spline Flow (NSF) is a density estimator that utilizes neural networks in order to transform a simple distribution (e.g. a typical Gaussian) into a more complex and better-fitting distribution on the parameters θ given x via monotonic and rational-quadratic splines, whose definitions were beyond the scope of our research. **Masked Autoregressive Flow (MAF)** is a conditional normalizing flow which is typically used as a default baseline flow for SBI-based networks, and as such is faster during training and computation. This comes at a trade-off of being more susceptible to underfitting than other flows.

3.2.3 Day 5

After performing 95 total runs, of both hyperparameter grid search and manual runs on certain architectures, the final best architecture used for all future work was the following

Input	Depth	Width	Act.	Flow	LR	Opt.	Batch	Epochs	Test RMSE
Non-PCA 110	3	16, 16, 8	ReLU	MAF	5.00×10^{-4}	Adam	64	86	0.1030

Table 5: The final SBI NPE architecture, used for all subsequent uncertainty quantification work. The normalizing flow was additionally calibrated to perform 8 transformations with 64 hidden features.

This model was deemed better than the previous grid-search one found in Day 4 as this was the best build overall that did not overfit to the training data. Additionally, the final models for each of my team members were also finalized as follows:

Model	Depth	Width	Act.	Loss	LR	Opt.	Batch	Test RMSE
Ensemble (5 nets)	3	[64, 256]	ReLU	MSE	5.00×10^{-4}	Adam	256	0.0953
Gaussian Predictor	3	128, 64, 2	ReLU	NLL	1.72×10^{-3}	AdamW	64	0.0903

Table 6: Final architectures for the Ensemble and Gaussian Predictor models. The Ensemble model had hidden layer widths vary between 64 and 256 neurons for each layer in each network in the ensemble, with the output layer outputting a single value representing the predicted metallicity.

However the goal of today was not to find better and better models for each of us, instead it was to discuss each of our final methodologies with Prof. Speagle and TA Ian Zhang, and to conceptually understand why each of our models work through rigorous mathematical discussions on the board. Through them, some previous misunderstandings about how my model, using SBI, were hastily clarified by Prof. Speagle. A few small, but incorrect assumptions—which aren't remembered anymore—about how the model worked in principle were fixed, and which are the ones outlined throughout the paper thus far. The insight was extraordinarily helpful as Prof. Speagle was the only one in the whole group who had worked with SBI before, or at least had knowledge of it, so it was very helpful to get a practical and mathematical intuition through a test of understanding through him.

Finally, the last portion of today was used to give a brief introduction to what uncertainty quantification means in different contexts and methods, where we will delve further in-depth in the next section.

3.3 Part 3: Uncertainty Quantification

The premise of uncertainty quantification is to produce actionable, valid insight as to how wide error bars are within a predictive model, whether it's a more advanced machine learning model or a simpler statistical model. The latter introduces the fundamentals of uncertainty quantification, through the definitions of aleatoric and epistemic uncertainty as already mentioned, along with the basic premise of categorizing the variance/standard deviation on the predicted values. This practice is especially simpler to understand with the distribution-based models in the Gaussian Predictor and SBI, however it is highly applicable to any machine learning setting, including the Ensemble model. In our more advanced machine learning context, we wished to produce **coverage intervals** that fit to a certain percent of our dataset, in order to have our models be able to explain this as much as possible.

Definition - Coverage Intervals

Coverage intervals are a very general and well-used topic in the field of statistics, similar to confidence intervals. They simply mean that for each prediction on a target variable, the predicted value will fall within some interval width a set percentage of the time.

In our case, the target variable is predicted $[Fe/H]$dex and our set percentage is 90%. This means that, theoretically and given infinite simulations, the expected amount of times our predicted value will fall within the width will converge 90%.

However, if we placed a desired coverage of 100%, this isn't at all helpful for our use-case because the only way to guarantee that our predictions fall into the width 100% of the time is to have a width that is the entire range of the dataset, or higher. That does not provide any helpful context as to how confident our model is. Conversely, a coverage percentage that is too low will produce interval widths that are also too narrow.

First, we gained intuition as to how uncertainty quantification even works in general through a volunteered presentation by another Ph.D student working with Prof. Speagle, Leo Murao Watson. Leo had introduced his work on uncertainty quantification, along with research done and key formulae that are important to this task. For sake of privacy as the research is still in an internal state and is not published, no details will be divulged, but it should be noted that lots of deliberation was had after the fact, so that our core understanding of methods in uncertainty quantification is ensured.

Instead, we will now move on to describing the topics that *are* publicly available, and were set up by Prof. Speagle. They came in two varieties in this project, **manual** and **automatic** methodologies, of which the former uses a **fudge factor** (FF) to adjust coverage interval widths via a manually-found FF c, and the latter utilizes **conformal prediction** to automatically perform the same task using a score function s_i for the i stars in the dataset.

Important Note - Validation/Holdout Dataset Necessity for Uncertainty Quantification

It should be noted that uncertainty quantification should generally only be done on the **validation/hold-out data**, so as to not introduce bias into the calculations. Since the models are already trained and fit for the training data, the residuals will be systemically smaller and will provide dishonest narrow coverage intervals. This is not an issue when using the validation data, as it has never been seen before by the model.

The reader should note that the entirety of **Part 3** spans across Days 8, 9, 10 (skipping 6 and 7 as they were the weekend and no official research was done during this time).

3.3.1 Manual Methodology (Fudge Factor)

At first, we attempted to visualize exactly what type of uncertainty our linear baseline had. Through continually adding more and more data to the test dataset, we found that RMSE was converging to a value of around 0.197. Initially, a large drop in the RMSE was noted, as expected. The higher RMSE at the start indicated high epistemic uncertainty. While the epistemic uncertainty decreased as RMSE got lower, the convergence in RMSE as more data was thrown at it showed higher aleatoric uncertainty signals—the sign of irreducible noise in the data itself. This noise, conceptually, can stem from photon noise in the telescope, or information that simply doesn't exist that would be useful to explain our target variable. With aleatoric uncertainty, it usually follows that a larger error bar is populated, while with epistemic uncertainty usually that simply is a matter of getting more data to help the model train better (particularly, one should aim to acquire more data from the relatively under-sampled portions of the pre-existing data).

In Prof. Speagle's earlier collaborative work, the paper found that Gaia's chemical information is informative for metal-rich stars (which are also the dominating portion of the dataset, as seen above through EDA) and is less informative for metal-poor stars (which are lesser in quantity, the relatively under-sampled portion of our dataset). This corroborated with future findings in our work later on as well, and is an excellent indicator of **heteroscedasticity** in our data. [1]

Definition - Heteroscedasticity

Heteroscedasticity is a fairly common topic in statistics, but is an important one for this research that should be reiterated.

In general, it means that the noise level in the dataset is not constant across the entire set, and will thus introduce a difference in residuals when testing on different subsets of the data.

This is typically seen in experiments as having the size of the residuals in a model grow proportionally to how extreme the data samples' values are in relation to the rest of the dataset. This is also seen in our case, where the more metal-poor stars have higher residuals overall, mainly because they represent a very small portion of the dataset compared to other star types.

Later in Part 3, we will deduce why and where this uncertainty is higher or lower, for each groups of stars in the dataset. Let us formally define what a fudge factor is, and what it does.

Definition — Fudge Factor

The **fudge factor** c is a scalar multiplier tuned on the calibration set to stretch or shrink the half-width of a coverage interval until empirical coverage hits the target level. The resulting interval is:

$$\hat{y} \pm c \cdot k \cdot \sigma$$

where $\hat{y}$ is the predicted [Fe/H] dex, $k \approx 1.645$ is the Gaussian z-score for 90% coverage, and σ is the residual scatter on the calibration set. σ is the standard deviation of errors in the Ensemble and Guassian Estimator models, while it is half of the 84th percentile minus the 16th percentile for the SBI model, which roughly equates to 68% of the data—the size of one standard deviation in a normal distribution—given that SBI produces non-normal distributions. Concretely, c is found by sweeping a grid and selecting the value where calibration coverage crosses the target empirical coverage—in our case, 90%. The chosen c is then applied unchanged to the test set, which is never used during tuning as per the validation necessity for tuning during uncertainty quantification.

As aforementioned, a key precondition to performing uncertainty quantification is to have a validation dataset. Out of the 30,000 randomly-sampled stars from the full dataset, we used a 60/20/20 train-test-validation split (thus 6,000 for validation). Using this validation set, we fed it through the trained model to produce residuals, and a standard deviation that summarizes how large the residuals typically are, on this unseen data.

After finding the above information, then we try to find an approximate fudge factor c that is able to find the desired calibration coverage, through the following definition.

Definition - Interval Band & Width

Given a predicted value $\hat{y}$ and a fudge factor c, the **interval band** is defined as:

$$[\,\hat{y} - c \cdot k \cdot \sigma,\;\; \hat{y} + c \cdot k \cdot \sigma\,]$$

where $k \approx 1.645$ is the Gaussian multiplier that brackets the middle 90% of a normal distribution, and σ is the model's residual scatter. The **interval width** is simply the total span of this band:

$$\text{width} = 2 \cdot c \cdot k \cdot \sigma$$

A coverage of 90% means that 90% of true $[Fe/H]$dex values fall inside their respective band. The **median interval width** across all stars is referred to as **sharpness**. A lower value indicates tighter, more informative intervals, provided coverage is still met.

For our methods in general, we ended up using a simple method of plotting a uniform distribution of 20 or more potential c values, and pick one that is just about enough to cover 90% of the data. For example, on the linear baseline, the following figure is how the plot looked like.

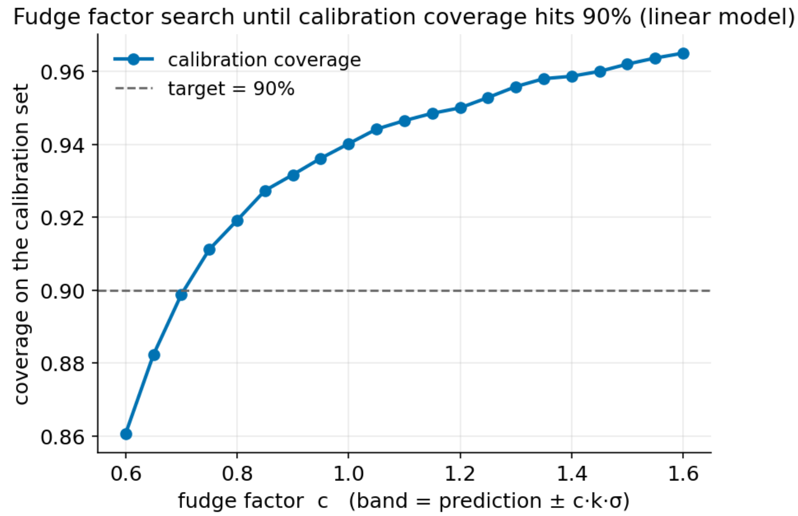

Figure 11: Found fudge factor $c \approx 0.71$ that enables coverage of about 90% of the data.

The above fudge factor is used on the entire dataset, thus it is meant to be an coverage interval meant to cover the data on **average** (or **marginally**). Something that might be more interesting is to see if this interval can also be used on conditional portions of the data. That is, on the different groups of stars present. We explored the empirical coverage for the surface gravity **log(g)** variable which describes the interval of giants and dwarfs, as well as the metallicity which describes the range of metal-poor to metal-rich stars. For example, in the linear baseline

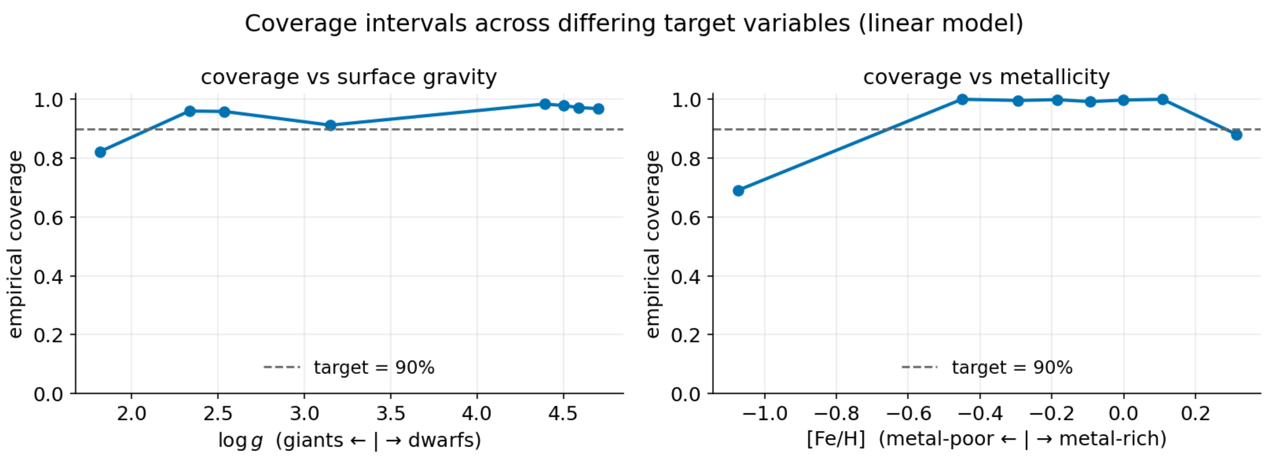

Figure 12: Overall, the empirical marginal coverage on on surface gravity is stronger across the entire range for $log(g)$ values, while there is lower conditional coverage for metal-poor stars. These metal-poor stars are, again, less common in the dataset compared to other star samples, which is correlated with having less conditional coverage on this subset of stars.

We noted earlier that the average/marginal coverage was good overall. However, with the conditional coverages on metallicity, the interval is under-covering luminous giants and over-covering the abundant metal-rich. This ties in to the earlier finding that our aleatoric uncertainty floor (the inability to go below a certain error rate) is heteroscedastic.

With all that in mind, we performed the same logic on each of our neural network models, with the plots for the SBI model shown below.

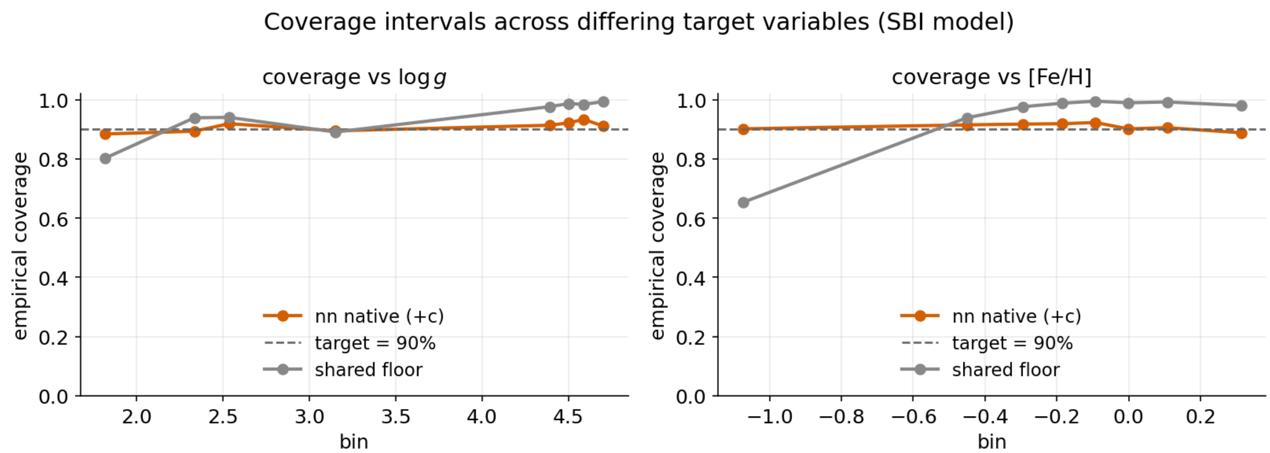

Figure 13: The found fudge factor value for the SBI model was $c = 1.10$, and the orange lines on the plot (nn native $(+c)$) showcase how the coverage is better aligned once the interval widths are adjusted from the shared floor.

The way to interpret how well a model is certain in this context is by having a c value that is closer to 1.0, which means it doesn't have to adjust the interval in order to cover the desired percentage of data. With a value of $c = 1.10$, it can be interpreted as needing to increase the interval width by about 10%. Additionally, the narrower a width is the better it is able to describe a particular subset of data, where the median of these widths is known as **sharpness**—the lower the sharpness the more certain a model is overall.

Note that the shared floor means having the same interval width for every single star, while the nn native interval widths are adjusted conditionally for each star. Thus, a sharper/narrower width is better as that means higher certainty in the prediction. Given the nn fudge factor made the interval widths narrower for metal-poor stars relative to the shared floor, this means the model is much more certain in predicting metallicity with this adjustment.

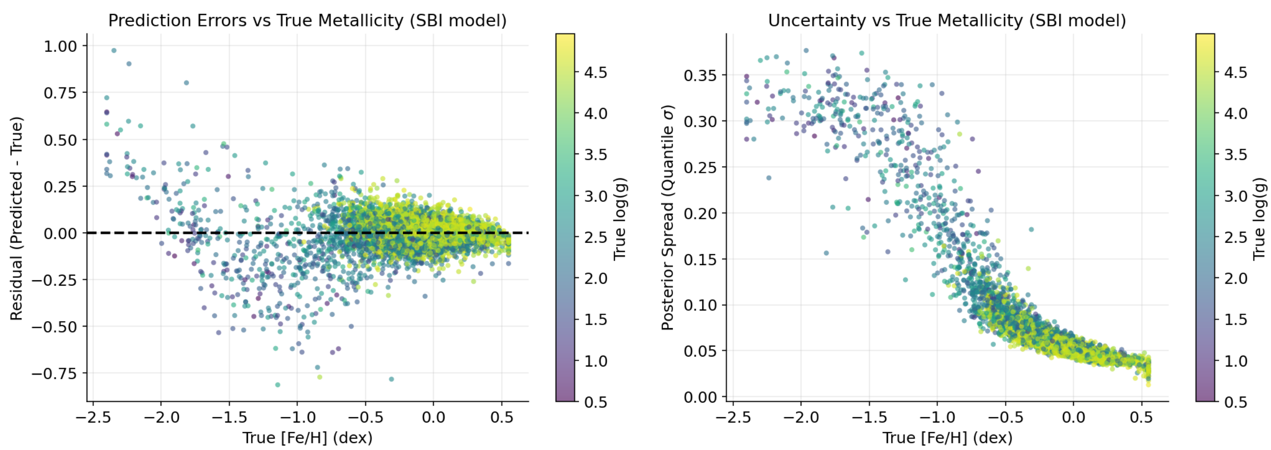

Figure 14: The left plot shows the residuals across the range of metallicity, which has a low overall residual for metal-rich stars and much higher and scattered for metal-poor stars. This is also backed-up by the right plot, which shows the uncertainty grows as the stars become more and more metal-poor. In terms of metrics, stars with $[Fe/H]\text{dex} \leq -1.5$ had an RMSE of 0.296 and median uncertainty of 0.316, while stars with $[Fe/H]\text{dex} \geq -1.5$ had an RMSE of 0.090 and median uncertainty of 0.058; a significant difference in favour of stars richer in metal.

From a glance, about 90% of the samples are clustered around a very low residual and uncertainty value, but then a sharp dip in accuracy is noted on metal poor stars. Given that there are so few of those stars compared to the metal-rich ones in the dataset, the model simply struggles at predicting metallicity due to an underlying amount of aleatoric uncertainty within the data, as noted by the rise in uncertainty in the right plot for those stars. Curiously, the model first underestimates the metallicity for the majority of the stars between metallicity ranges of $[-2.0, -1.0]$, and then suddenly overestimates below a metallicity of -2.0.

Nonetheless, this sets a strong framework for how uncertainty can be quantified. The main takeaway is that a fudge factor c that is closer to the value 1.0 will show that the model is quite certain in its predictions to cover some percentage of the dataset.

The reported fudge factor values for each of the models, including the SBI one, is as follows:

Model	**Fudge Factor** c	**FF Median Width**	**Conformal Median Width**
Linear Baseline	0.71	0.474 dex	0.469 dex
Umang (Ensemble)	3.90	0.206 dex	0.259 dex
Johan (Gaussian Predictor)	1.00	0.168 dex	0.171 dex
Edward (SBI NPE)	1.10	0.211 dex	0.265 dex

Table 7: Fudge factors and associated median interval widths found for each model. The linear baseline has no native FF interval width as its band is a constant shared floor; its conformal $\hat{q}$ of 0.235 dex serves as the direct comparison.

3.3.2 Automatic Methodology (Conformal Prediction)

Given our findings through the manual method of uncertainty quantification, the following task was to compare our manual method with the automatic method to quantify our uncertainty. The following presents the use of **conformal prediction** on each of our models.

Definition - Conformal Prediction

Rather than manually tuning a fudge factor c to hit a target coverage, **conformal prediction** automates this by computing a calibration quantile $\hat{q}$ directly from the calibration set's residuals, using a **non-conformity score function** that measures how surprising each star's true $[Fe/H]$dex is relative to the model's prediction. The score function can be as simple as the absolute error $s_i = |y_i - \hat{y}_i|$, or as complex as necessary depending on the particular problem the researcher/user attempts to solve. The resulting half-width for every interval is:

$$\hat{q} = \text{Quantile}_{1-\alpha}(\{s_i\}_{i=1}^{n_{\text{cal}}})$$

so that the coverage band $\hat{y} \pm \hat{q}$ upholds a finite-sample **guarantee**:

$$\mathbb{P}(y_{n+1} \in C_\alpha(x_{n+1})) \geq 1 - \alpha$$

provided the calibration and test stars are **exchangeable**, meaning they drawn from the same distribution in no fixed order. Conformal prediction assumes no shape for the error distribution, thus it is **distribution-free** and uses quantiles rather than a Gaussian multiplier. It is also **model-agnostic**, so it works for any type of model no matter the underlying output shape. [3, 4]

In essence, conformal prediction performs the same calibration coverage method as we did by manually finding a fudge factor, just automating it through minimizing a non-conformity score function and creating the intervals from a calibration quantile based off of the score function output distribution. The reason that this method works at all is due to the assumption of **exchangeability** in the data, meaning that stars are drawn from the same dataset in no defined order, thus having each new star's error being statistically considered as one more calibration error. Should there be any testing split that is predominantly one class of stars (e.g. dwarfs) and then conformal prediction is performed on a subset of other stars, this assumption is broken and conformal prediction will generate false results. This effectively **ranks** each calibration score, and thus every star, based on how accurate (or certain) the model is in predicting the new data. Hence, conformal prediction is a **distribution-free** method given that it assumes nothing about the error distribution's shape (further shown through the use of quantiles rather than, for example, some Gaussian multiplier) and is also **model-agnostic**. These parameters made it an effective choice to compare each of the proposed prediction models.

Of course, conformal prediction also requires a hold-out/validation dataset to use in order to perform this uncertainty quantification. Through using an absolute error score function $s_i = |y_i - \hat{y}_i|$ on the linear baseline to find a marginal coverage of $\sim 90\%$, a value of $\hat{q} = 0.235$ dex is found, which determines that the half-width of every coverage interval is 0.235 dex through this method. Empirically, this means that $\hat{q}$ is the value where about 90% of the metallicity error is no larger than it. This is not a conditional coverage, but marginal, thus it is sensitive to the same outlying stars in the dataset as the manual method was. However, later we will note a method of performing conformal prediction to find conditional coverage instead.

Astutely, this is exactly comparable with the coverage interval width that is found with the fudge factor method, wherein a fudge factor c also aims to minimize these widths. The same logic applies for conformal prediction, where the score function to find some guaranteed marginal coverage. The one downside to conformal prediction is that conformal bands will have a constant width given the restrictions upon the prior assumption of it being distribution-free. This usually implies that sharpness performed through conformal prediction will be higher, unless a strong score function is created that matches the residual distribution well. The upside, however, is that a set coverage percentage (in our case, 90%) of future predictions are guaranteed to fall within the coverage interval (the predicted range), whereas fudge factor methods do not have this guarantee and could fail on new data. As such, models that utilize conformal prediction can be considered statistically viable assuming the sharpness and coverage match the user's needs.

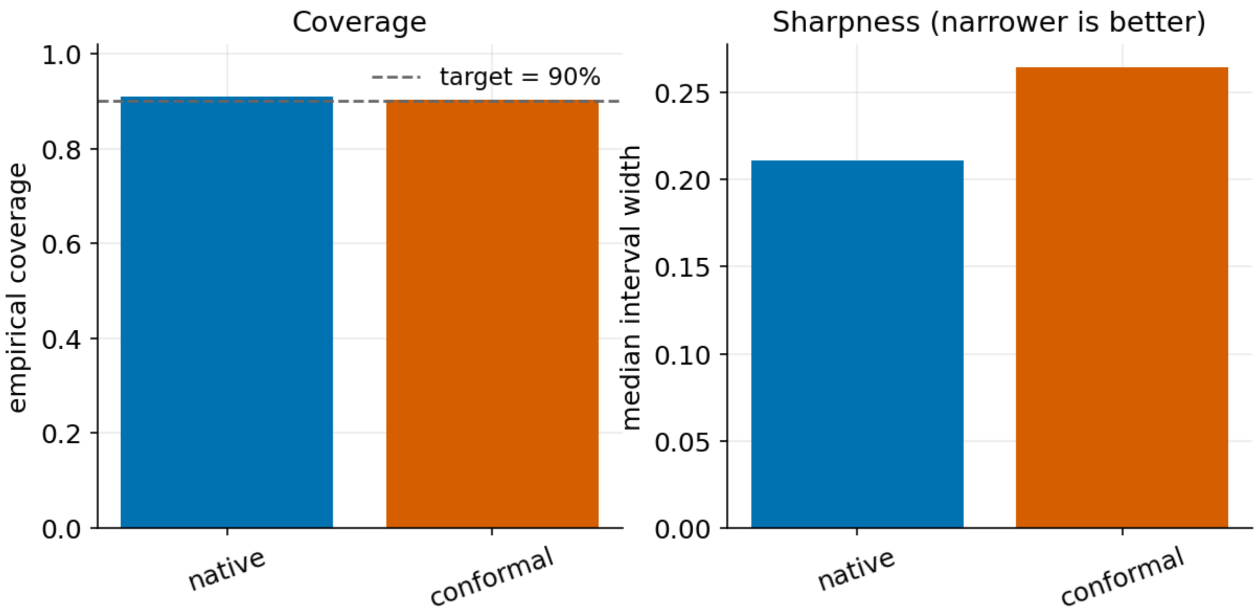

Figure 15: This shows the difference in sharpness that nn native widths have in comparison to conformal prediction widths. Since sharpness is lower in the nn native widths compared to conformal prediction, the native intervals are tighter, though conformal prediction carries the formal coverage guarantee.

Another limitation of conformal prediction is that it produces guaranteed *marginal* coverage, thus for any particular groups of stars where the conditional coverage is poor (e.g. the metal-poor stars), then conformal prediction falters in guaranteeing coverage.

One key consideration with this method in our work is the definition of the scoring function s_i. While the Ensemble and SBI models simply used the base absolute error score function to minimize, the Gaussian Predictor model was tuned on minimizing the z-score of each prediction. With further development and time, some stronger scoring functions that better suited this problem may have been defined. At the time of performing conformal prediction, some more research and insight was needed into understanding how it worked theoretically, and thus left little time to consider different scoring functions; a possible future improvement.

There do exist methods to expand conformal prediction to a more robust and accurate level. These methods are Conformalized Quantile Regression (CQR), Mondrian (group-conditional) conformals, and weighted conformals. Unfortunately, the group was not able to explore these extensions in depth as we were attempting to wrap our heads around how conformal prediction works in general, and needed to create the poster in due time.

Definitions - Extensions to Conformal Prediction

Three principled extensions address the limitations of standard (marginal) conformal prediction:

Conformalized Quantile Regression (CQR) resolves the tension between the conformal guarantee and adaptive widths. Rather than applying a single constant $\hat{q}$ to a point prediction, CQR starts from a model that already predicts a low quantile and a high quantile for each star. The non-conformity score measures how far outside this raw band the true $[Fe/H]$dex falls, and a single conformal correction $\hat{q}$ nudges both edges outward. The result is an interval that *adapts* in width per star, with wider widths for uncertain metal-poor stars, and narrower widths for confident metal-rich ones. These widths are adjusted in such a way to continue retaining the marginal coverage guarantee.

Mondrian (Group-Conditional) Conformal achieves conditional coverage over defined groups by running the entire conformal recipe *separately* within each group. In our case, splitting calibration stars into giants and dwarfs and computing a separate $\hat{q}$ for each means the giants' band is sized by giants' own residuals, guaranteeing coverage holds within each bin rather than only on average.

Weighted Conformal Prediction is the principled fix for covariate shift — when the test stars are drawn from a different distribution than the calibration stars. Rather than treating all calibration scores equally, each score is reweighted by the ratio of the test density to the calibration density ($w_i \propto p_{\text{test}}(x_i)/p_{\text{cal}}(x_i)$), and the weighted quantile is taken. This restores the coverage guarantee when the standard exchangeability assumption is violated.

It is encouraged to those curious (including the group) to look into these topics and see what value they can bring.

Despite any shortcomings that may come from both methods, they both carry significance in producing viable models through analysis and seeing how well the model, well, models the data in a general manner. The main takeaways of conformal prediction is that it is powerful for large-scale data and ensuring label transfer at scale. Conformal intervals produce a guaranteed, trustworthy estimate for large amounts of data.

3.4 Part 4: Results, Poster Creation and Project Acknowledgements

3.4.1 Results and Findings

The final numerical results summarizing our findings throughout the entire process are as follows:

Model	RMSE	R^2	FF c	FF Width	Conf. Width
Linear Baseline	0.198	0.788	0.71	0.474 dex	0.469 dex
Ensemble	0.095	0.948	3.90	0.206 dex	0.259 dex
Gaussian Prediction	0.087	0.960	1.00	0.168 dex	0.171 dex
Simulation-Based Inference	0.103	0.943	1.10	0.211 dex	0.265 dex

Table 8: Model comparison. RMSE and R^2 are on the test set. c is the fudge factor. FF Width and Conf. Width are the fudge factor and conformal median interval widths respectively.

Without having a formal hypothesis in mind as we were researching and building models, we deduced that more complex models were certainly required to appropriately train for accurate predictions in this particular use-case. Within each of these models, gathering a deep intuition as to how the model worked for this particular setting was key for appropriately tuning the model, taking into consideration the effects of aleatoric and epistemic uncertainty, which affected each model differently. After viewing the results, it was clear that our scoring function needed to be developed further in order to produce stronger metrics based on a better-fitting score function that minimizes interval widths appropriately. Nonetheless, the Gaussian Predictor model had a fairly well-tuned score function that had very minimal conformal interval width increase relative to the other methods. Ideally, conformal prediction would serve to *decrease* the widths, especially since it is an automatic method that has the potential to be used in large-scale systems that require automation. However, this is an improvement for the team to consider for future work utilizing conformal prediction.

Nevertheless, the team also observed that these widths were hitting a floor, much like our RMSE and R^2 scores from before. This signified that there was underlying aleatoric uncertainty that kept our models from further accurately producing predictions. The increases in interval widths for the Ensemble and SBI going from fudge factor to conformal widths also signified the presence of stronger epistemic uncertainty, especially relative to the Gaussian Predictor model. This epistemic uncertainty came from two aspects. The first being conformal prediction scoring function not being necessarily appropriate for the model, which led to those larger widths going from fudge factor to conformal widths. The second being uncertainty inside the models themselves, noted by the fudge factor c value not being equal to 1, which meant that some level of adjustment was required to ensure the model has better coverage on the data. Both variations of epistemic uncertainty, in fact most epistemic uncertainty causes in general, can be mitigated by having better-tuned models and assumptions. With aleatoric uncertainty, that is simply an issue with data quality and is harder to mitigate.

With that said, this paper has repeatedly mentioned the correct assumption that a Gaussian can appropriately model the output distribution, and the results show that this was a correct assumption. The fudge factor c value for the Gaussian Predictor was 1.00, signifying negligible adjustment necessary to the widths to produce appropriate coverage on the data. Additionally, a Q-Q plot (made in the later stages of our work) created confirmed that the theoretical distribution is indeed a normal distribution.

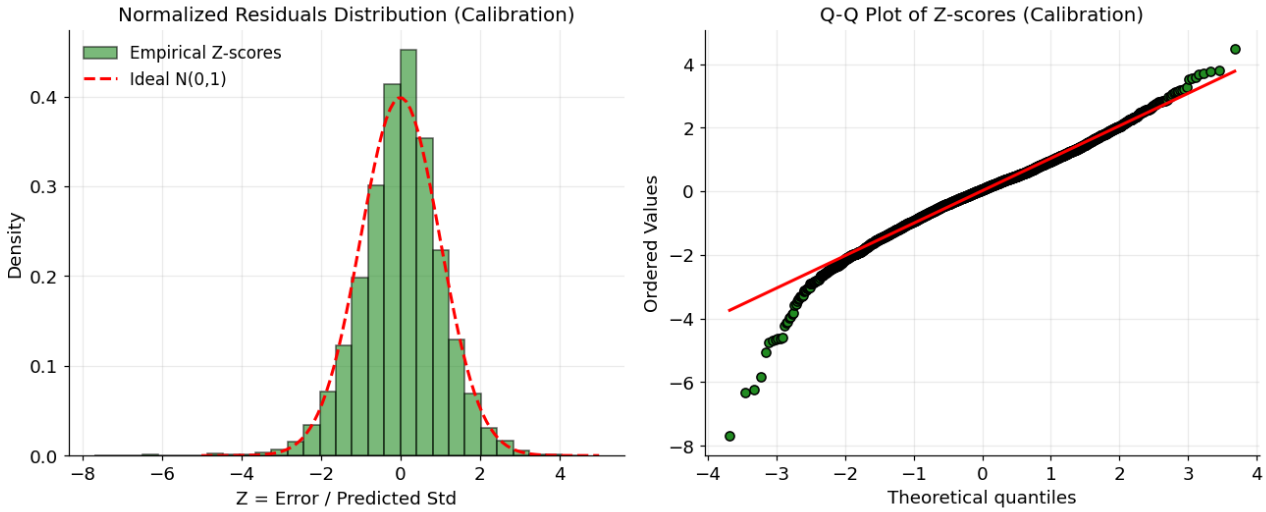

Figure 16: Left plot shows that the residual errors followed a normal distribution, which led to the z-score scoring function for the Gaussian Predictor's conformal prediction methods. The Q-Q plot showed that overall the z-scores of the errors on the calibration set are very close to normal for the majority of the data.

The overarching purpose of this project was to deduce and learn and apply about novel methods (to the undergraduate student researchers) surrounding uncertainty quantification, and also to extend our model-building capabilities. The latter was well-achieved by each of us, as we had all learned a novel, complex method of using neural networks and taught each other how each of them worked. As for the former, we had also learned manual and automatic methodologies of how uncertainty can be quantified in machine learning models. We are quite proud of our efforts and have learned a great deal throughout the two weeks of research undertaken.

3.4.2 Poster Creation and Presentations

The research portion of Day 11 of the program, was kept as the time to produce a viable academic poster for presenting our work in a lightning-talk style on the following day. However, this had come as a little bit of a surprise for our group as we weren't aware (completely out of our own fault) that the posters were actually due at 4:00 PM and not when the research portion ends at 5:30 PM, let alone the 11:59 PM the majority of university students are familiar with. At the time of just after 3:00 PM we had looked a little closer at the message regarding the deadlines and realized that we had under an hour to create a professional academic poster. Anyone who has already gone through a poster-making process before knows that this is nigh on a tall task and that anything done essentially from start-to-finish in an hour is practically impossible. However, we set the expectations aside and dove in with our conviction!

And so, within about 40 minutes, thanks in part to our many hours of questions asked post-research times amongst each other thus generating a strong understanding of exactly what it is we did—but mainly TA Ian's unfaltering hypersonic speed at which he was able to compile our assortment of information, numbers, and graphs precariously placed onto a Google Doc—a fully-completed poster was submitted with a few minutes to spare.

It only served as huge irony that we were in fact the very first group to submit a poster and had learned the next day that some others had managed to get quite long extensions to their submissions—hours more, in fact. Nonetheless, we remained proud of our finished product despite the small errors that we hadn't been able to fix in time to meet the deadline.

Over the course of the final two and a half hours on Day 12, we had presented our project's purpose, methodologies, findings, and conclusions multiple times to fellow UTSSRP students, faculty members of the Department of Statistics, Ph.D students of the same department, and more. Presentations were of about 3 minutes in length, but not at all formally timed. Ad-hoc Q&A was also part of the process as everyone in the setting had natural curiosity to explore everyone else's project, many topics are niche enough that there's always something new to be learned! Faculty members were especially in-depth with their questions, hence the presentations were the culmination of not only our project's work and depth, but also a showcase of how we were able to integrate different lectures and seminars into our presentation and Q&A as well. It was a great experience, which finished with a series of group photos, closing words, and many goodbyes and thanks to the faculty.

4 Acknowledgements

As a fabulous 2 weeks blitzed by at the University of Toronto St. George campus, it was a fantastic experience that I highly recommend to any and all undergraduates to pursue if they can. Especially so if they are interested in looking to further their statistical mathematics skills, be open to learning some newer concepts that will enhance their current arsenal, and to gain the opportunity to have a connections to a very strong network of highly successful statisticians. The program seeks to find the next top talent that could walk the Department's halls one day as a researcher and graduate student, while also granting the opportunity for everyone to grow into a highly capable individual, extending a vast amount of learning in just two weeks. It was a pleasure to be a part of, and wish everyone involved all the best.

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Me and my teammates would like to thank Prof. Joshua S. Speagle for setting up the Jupyter environment and notebooks for us in such a way that enabled exploring these topics extremely efficiently. His efforts allowed us to spend minimal time on coding specifics, and more to actually understand the problem at hand, the physics and statistics behind it, and to learn new methodologies in uncertainty quantification for each of us. Additionally, thank you to TA Ian Zhang for always being there to answer our questions, supporting us, teaching us, bringing us many good laughs, and to help us with the final poster push in such a short amount of time. And finally, thank you to Leo Murao Watson for coming in while we were learning about conformal prediction and providing us with a detailed presentation behind the theory of conformal prediction so that we better understood it.

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- Prof. Meredith Franklin, Professor of the University of Toronto Department of Statistical Sciences
- Prof. Stanislav Volgushev, Associate Professor of the University of Toronto Department of Statistical Sciences
- Prof. Joshua S. Speagle, Assistant Professor of the University of Toronto Department of Statistical Sciences, and Supervisor for the Uncertainty Quantification for Predicting Stellar Properties from Gaia Spectra project
- Prof. Ricardo Baptista, Assistant Professor of the University of Toronto Department of Statistical Sciences
- Prof. Elena Tuzhilina, Assistant Professor of the University of Toronto Department of Statistical Sciences

- Ian Zhang, Ph.D of Statistics Student supervised by Prof. Joshua S. Speagle in the University of Toronto Department of Statistical Sciences, and TA for the Uncertainty Quantification for Predicting Stellar Properties from Gaia Spectra project
- Leo Murao Watson, Ph.D of Statistics Student supervised by Prof. Joshua S. Speagle and Prof. Radu Craiu in the University of Toronto Department of Statistical Sciences, and presented "An analysis of Bayesian Optimization with Conformal Prediction Sets"
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- Johan Phan, B.Sc Computer Science and Mathematics Student of the University of California, Berkeley and team member for the Uncertainty Quantification for Predicting Stellar Properties from Gaia Spectra project

Lecturers (in order of delivery date)

- Prof. Gracia Dong, Assistant Professor in the Teaching Stream of the University of Toronto Department of Statistical Sciences
- Max Ma, Lecturer of the University of Toronto Master of Financial Insurance (MFI) program
- Prof. Keith Knight, Professor of the University of Toronto Department of Statistical Sciences
- Prof. Elena Tuzhilina, Assistant Professor of the University of Toronto Department of Statistical Sciences
- Prof. Ricardo Baptista, Assistant Professor of the University of Toronto Department of Statistical Sciences
- Prof. Joshua S. Speagle, Assistant Professor of the University of Toronto Department of Statistical Sciences
- Prof. Sebastian Jaimungal, Department Chair and Professor of the University of Toronto Department of Statistical Sciences, and Head of the UTSSRP

Seminar Panelists (in order of delivery date)

- Prof. Jun Young Park, Assistant Professor of the University of Toronto Department of Statistical Sciences
- Prof. Thibault Randrianarisoa, Assistant Professor of the University of Toronto Department of Statistical Sciences
- Prof. Nancy Reid, Fellow of the Royal Society of Canada, Canada Research Chair for NSERC, Fellow of the Institute of Mathematical Statistics, and University Professor Emeritus of the University of Toronto Department of Statistical Sciences
- Prof. Mufan Li, Assistant Professor of the University of Waterloo Department of Statistics and Actuarial Science
- Kevin Zhang, Ph.D of Statistics and Research Machine Learning Scientist II at TD Canada
- Prof. Michael Moon, Assistant Professor in the Teaching Stream of the University of Toronto Department of Statistical Sciences
- Prof. Jeff Rosenthal, Fellow of the Royal Society of Canada, Fellow of the Institute of Mathematical Statistics, and Professor of the University of Toronto Department of Statistical Sciences
- Prof. Meredith Franklin, Professor of the University of Toronto Department of Statistical Sciences
- Prof. Silvana Pesenti, Associate Professor of the University of Toronto Department of Statistical Sciences

Social Committee and Admin

- Kal Romain, Manager of Media & Communications of the University of Toronto Department of Statistical Sciences, and UTSSRP Administrator
- Kachely Peters, Social Media and Event Assistant of the University of Toronto Department of Statistical Sciences
- Mandy Yao, Ph.D of Statistics Student supervised by Prof. Meredith Franklin of the University of Toronto Department of Statistical Sciences

- Bernard Miskic, Ph.D of Statistics Student supervised by Prof. Meredith Franklin and Prof. Patrick Brown of the University of Toronto Department of Statistical Sciences
- Arian Hashemzadeh Amirkhizi, Ph.D of Statistics Student supervised by Prof. Elena Tuzhilina and Prof. Piotr Zwiernik of the University of Toronto Department of Statistical Sciences
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The entire program would not have been possible without the immense amount of logistics and planning that had gone behind the organization of it. Each individual listed, and many more, ensured that the UTSSRP was a big success, and I would like to thank everyone involved making this immense program a reality. I have since added a Minor in Statistics designation to my degree plan following the UTSSRP, in big part due to the profound effect and spike in curiosity in statistics the program brought to me.

Finally, one last thank you to all of the other students who joined the esteemed cohort of the UTSSRP this year. Your enthusiasm showed throughout these few weeks, and I was very glad to have met all of you. Very bright futures ahead for each of us, and am excited to see where our paths take us.

Many big thanks once again, and I wish all the best to the future cohorts of the UTSSRP.

- Edward Tanurkov, Bachelor of Computing (Honours) Student of Queen's University

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